We shall deal with positive form as linear operator at first, and we will generalize.

Def. A linear operator $T: V \to V$ on a fin. dim inn. prod space V is called positive if:

T is Hermitian and $\langle T(\alpha), \alpha \rangle > 0$ for all $\alpha \neq 0$

Prop. Let T, U be Hermitian on fin. dim inn. prod space V.
And, $A = [T]_\beta$, where β is any orthonormal basis for V.

1). T is positive if and only if all eigenvalues of T are positive.

2). T is positive if and only if $\sum_{i,j} A_{ij} a_j \overline{a_i} > 0$ for all nonzero tuple $(a_1, \cdots, a_n)$.

3). T is positive if and only if

$$A = B^* B \quad \text{for some } B.$$

Proof. Denote $\beta = \{\alpha_1, \cdots, \alpha_n\}$. Let $B = [I]_\beta^{\{e_1, \cdots, e_n\}} = (\alpha_1 \ \alpha_2 \ \cdots \ \alpha_n)$,
And, let $\beta' = \{\beta_1, \cdots, \beta_n\}$ be an orthonormal basis for V consisting eigenvectors of T.

1). If T is positive, then $\langle T(\beta_i), \beta_i \rangle = \langle c_i \beta_i, \beta_i \rangle = c_i > 0$, where c_i is eigenvalue.
Conversely, given $X \in V$, $X = x_1 \beta_1 + \cdots + x_n \beta_n = \langle X, \beta_1 \rangle \cdot \beta_1 + \cdots + \langle X, \beta_n \rangle \cdot \beta_n$, so

$$\langle T(X), X \rangle = \left\langle \langle X, \beta_1 \rangle \cdot c_1 \cdot \beta_1 + \cdots + \langle X, \beta_n \rangle \cdot c_n \beta_n, \langle X, \beta_1 \rangle \cdot \beta_1 + \cdots + \langle X, \beta_n \rangle \cdot \beta_n \right\rangle$$

$$= \sum_{i=1}^{n} c_i \cdot |\langle X, \beta_i \rangle|^2 > 0 \quad \text{if } X \neq 0 \text{ and } c_i > 0.$$

2). Given $X \in V$ s.t. $[X]_\beta = (a_1, \cdots, a_n)$,

$$\langle T(X), X \rangle = \left\langle \sum_{j=1}^{n} a_j T(\alpha_j), \sum_{i=1}^{n} a_i \cdot \alpha_i \right\rangle = \sum_{j=1}^{n} \sum_{i=1}^{n} \langle T(\alpha_j), \alpha_i \rangle \cdot a_j \cdot \overline{a_i},$$ so Proved.

(+Recall: $A_{ij} = \langle T(\alpha_j), \alpha_i \rangle$).

3). If $A = B^* B$, then $\langle B^* B x, x \rangle = \langle Bx, Bx \rangle = \|Bx\|^2 > 0$. Conversely
since $[T]_{\beta'} = D$ is diagonal with positive values, where

$$D = \begin{pmatrix} c_1 & & 0 \\ & c_2 & \\ & & \ddots \\ 0 & & & c_n \end{pmatrix} \quad c_i\text{'s are eigenvalues of } T$$

so we can define $\sqrt{D} = \begin{pmatrix} \sqrt{c_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{c_n} \end{pmatrix}$. Now, for $P = [I]_{\beta'}^{\beta}$,

$$A = P^{-1} D P = P^{-1} \cdot \sqrt{D} \cdot \sqrt{D} \cdot P = \left(\sqrt{D} P\right)^* \cdot \left(\sqrt{D} P\right), \text{ since } P \text{ is unitary, } \sqrt{D} \text{ is symmetric.}$$

And, there are more interesting problems,

4). If T and U are positive s.t $T^2 = U^2$, then $T = U$.

5). If T and U are positive s.t $TU = UT$, then TU is positive.

Proof. Let $\beta = \{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for V consisting eigenvectors of T. Let $c_1, \cdots, c_n$ be eigenvalues corresponding to $\alpha_1, \cdots, \alpha_n$, respectively. ∎ setting For each $1 \le i \le n$,

$$T^2(\alpha_i) = c_i^2 \cdot \alpha_i = U^2(\alpha_i).$$

And, $c_i^2 \alpha_i = U^2(\alpha_i) \Leftrightarrow (U^2 - c_i I)(\alpha_i) = 0.$

$$\Leftrightarrow (U + c_i I) \circ (U - c_i I)(\alpha_i) = 0.$$

If $V = (U - c_i I)(\alpha_i) \neq 0$, then

$$U(V) = -c_i V$$

So V is an eigenvector of U with negative eigenvalue, it is contradiction to U is positive, so $U(\alpha_i) = c_i \alpha_i$ for all $1 \le i \le n$. The linear map is determined by maps basis to some vectors, so $T = U$ ∎

Since T and U commute and both are Hermitian,

$$(TU)^* = U^* T^* = UT = TU$$

So TU is Hermitian. For the above β, define a linear map as

$$\sqrt{T}(\alpha_i) = \sqrt{c_i} \cdot \alpha_i.$$

Then, for each $1 \le i \le n$,

$$U\sqrt{T}(\alpha_i) = U(\sqrt{c_i}\alpha_i) = \frac{1}{\sqrt{c_i}} \cdot U(c_i \alpha_i) = \frac{1}{\sqrt{c_i}} \cdot U(T(\alpha_i)) = \frac{1}{\sqrt{c_i}} \cdot TU(\alpha_i)$$

Def. A form f on an inn. prod space V is called non-negative if:

f is Hermitian and $f(\alpha,\alpha)\geq 0$ for all $\alpha\in V$.

and, if $f(\alpha,\alpha)>0$, $\alpha\neq 0$, it is called positive,

Example. Let f be a form on fin. dim inn. prod space V, let $\mathcal{B}=\{\alpha_1,\cdots,\alpha_n\}$ be an ordered basis.
Let A be a matrix of f in $\mathcal{B}$. That is, $A_{jk}=f(\alpha_k,\alpha_j)$

If $\alpha=x_1\alpha_1+\cdots+x_n\alpha_n$, then

$$f(\alpha,\alpha)=f\left(\sum_j x_j\alpha_j,\sum_k x_k\alpha_k\right)=\sum_j\sum_k x_j\overline{x_k}\cdot f(\alpha_j,\alpha_k)=\sum_j\sum_k A_{kj}\cdot x_j\cdot\overline{x_k}.$$

So, f is non-negative iff and only if $A=A^*$ and $\sum_j\sum_k A_{kj}x_j\overline{x_k}\geq 0$, for all x_j,x_k
Equivalently, f is non-negative if and only if

$$g:F^n\times F^n\to F:(X,Y)\mapsto Y^*AX$$

is non-negative form, $F=\mathbb{C}$ or $\mathbb{R}$,

Thm. Let $F=\mathbb{C}$ or $\mathbb{R}$, and $A\in M_{n\times n}(F)$.
A function $g:F^n\times F^n\to F:(X,Y)\mapsto Y^*AX$

is a positive form if and only if $A=P^*P$ for some invertible $P\in M_{n\times n}(F)$,

Proof Suppose that g is positive form. Then, it is an inner product on F^n,
So there is an orthonormal basis $\{Q_1,\cdots,Q_n\}$ for F^n, satisfying

$$g(Q_i,Q_j)=Q_j^*AQ_i=\delta_{i,j}.$$

So, take $P=\begin{pmatrix}Q_1 & \cdots & Q_n\end{pmatrix}^{-1}$, then $PAP^*=I$, or $A=P^*P$.

Conversely, if $A=P^*P$ for some invertible P, then $A^*=(P^*P)^*=P^*P=A$ and

$$X^*AX=X^*P^*PX=(PX)^*\cdot(PX)>0 \text{ if } X\neq 0.$$

So proved.

Def. Let $A \in M_{n\times n}(F)$. The principal minors of A are:

$$\Delta_k(A) \overset{\text{def}}{=} \det \begin{bmatrix} A_{11} & \cdots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \cdots & A_{kk} \end{bmatrix} \qquad 1 \le k \le n$$

Lemma. Let $A \in M_{n\times n}(F)$ be an invertible. TFAE:

a) There is a uni-upper triangle matrix P such that AP is lower triangular.

b) The principal minors of A are all different from 0.

Pf. Note: B is upper triangle iff $B_{ik} = 0$ if $i > k$.

Suppose that a) holds.

Denote $A_1, \cdots, A_n$ are columns of A, and $B_1, \cdots, B_n$ are column of $B = AP$.

Then, $B_1 = A_1$ by P is uni-upper, and

$$B_r = A_r + \sum_{j=1}^{r-1} P_{jr} \cdot A_j$$

(observe that

$$(B_1 \cdots B_n) = (A_1 \cdots A_n) \cdot \left(I + \begin{pmatrix} 0 & & P_{1k} \\ & \ddots & \\ 0 & & 0 \end{pmatrix} \right) \qquad (P_{ik} = 0 \text{ if } i \ge k)$$

$$= (A_1 \cdots A_n) + \left(\underbrace{\sum_{j=1}^{r-1} P_{jr} \cdot A_j}_{r\text{'th column}} \right)$$

).

Therefore, for each $1 \le k \le n$,

$\begin{bmatrix} B_{11} & \cdots & B_{1k} \\ \vdots & \ddots & \vdots \\ B_{k1} & \cdots & B_{kk} \end{bmatrix}$ is obtained by adding to the r th column of

$\begin{bmatrix} A_{11} & \cdots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \cdots & A_{kk} \end{bmatrix}$ as a linear combination of its other columns.

This operation cannot change the determinant, so, $\Delta_k(A) = \Delta_k(B)$.

And, since A, P are invertible, so is B, thus $\Delta_n(B) = \det B \neq 0$.

Now, $\Delta_k(B) = B_{11} \cdots B_{kk} \neq 0$, since it is triangular.